

A Short Note on Transcript Attacks on Lattice-based Signature Schemes based on the Round-Off Algorithm

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Abstract

This short note expositis the work of Nguyen and Regev in [5]. In this paper the authors devise an attack on hash-and-sign schemes that implicitly solves the closest vector problem using the round-off algorithm. We discuss the proposed algortihm that leaks vectors in the secret basis if it is given access to a transcript of signatures. Particularly, we construct the fourth moment operator of the singature samples which encodes the secret basis as its points of minima.

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1 Introduction

In recent years many cryptographic frameworks or schemes have been proposed which exploit the hardness of lattice problems – GGH, NTRU to name a few ([2],[3]). Such schemes come under a hash-and-sign paradigm, where a document is hashed to the ambient space of a certain high dimensional

lattice, and the signature (or encryption) is computed as the vector that joins this point to a lattice point very close to it. The signatures generated by these schemes therefore lie in a parallelepiped generated by these basis vectors of the lattice. The quality of the signature is a function of its size – shorter the better – which, in turn, is a function of the quality of the secret basis. That is, more orthogonal bases lead to parallelepipeds of smaller volume, and hence, shorter signatures.

Naively forging a signature requires an attacker to solve the closest vector problem. This is very difficult without access to the secret basis. However, a transcript of many signatures leaks additional information. If the signatures are uniformly distributed in the aforementioned parallelepiped, a sufficient number of samples can leak the shape of the secret basis. Nguyen and Regev propose a novel algorithm in [5] to achieve this specific task. Their ingenuity lies in the construction of a moment operator (as opposed to fixed moments) which encodes the secret basis as its points of minima.

In the coming sections a vector x will always have coordinates x_i s and a matrix B will have rows b_i s, unless specified otherwise.

2 The Hash-and-Sign Paradigm

In this section we will formally define the Hash-and-Sign paradigm which we briefly glanced at in the introduction. Let $\mathcal{M}$ be the space from which we draw the messages or documents. Then the signer is equipped with a hash function $H: \mathcal{M} \rightarrow \mathbb{R}^n$. Further, in $\mathbb{R}^n$ sits a lattice $\Lambda = \mathcal{L}(B)$, where Λ is the integral linear span of the rows of a nonsingular matrix B .

Given a target point $t = H(m) \in \mathbb{R}^n$ obtained from a message m , the signer attempts to find a lattice point $z \in \Lambda$ such that $\|t - z\|$ is small. One can use the round off method to compute such a vector z . For this one writes the vector t in the basis B as tB^{-1} , and then simply rounds each coordinate to the nearest integer to obtain the integral vector $\lfloor tB^{-1} \rfloor$. Finally the change of basis is undone to compute z as $z = \lfloor tB^{-1} \rfloor B$.

The signature for the message m is then just the vector $s = z - t$. One observes here that¹

$$s = z - t = (\lfloor tB^{-1} \rfloor - tB^{-1}) B \in \left[-\frac{1}{2}, \frac{1}{2}\right]^n B.$$

1. In the equation the notation $\left[-\frac{1}{2}, \frac{1}{2}\right]^n B$ is for the set $\{x_1 b_1 + \dots + x_n b_n \mid -\frac{1}{2} \leq x_j \leq \frac{1}{2}\}$.

From this we note that shortness of s is dependent on the geometry of B . In particular, the signature s belongs to a unit² parallelepiped generated by rows of B . If this parallelepiped has a small volume, it enforces shorter signatures. This is the key idea behind the construction of the signature scheme. We assume that the signer holds a basis B of the lattice Λ whose corresponding parallelepiped is small, that is, it is generated by fairly orthogonal vectors. The security of the signature scheme is therefore founded on the difficulty of finding such a basis.

Now a verifier has another lattice basis A , but this basis is ill conditioned, that is, the corresponding parallelepiped is far from a cuboid. Given a signature s , the verifier can verify that s is short, and that $H(m) + s$ (which is implicitly z if the signature is correct) is indeed a lattice point. However, without knowledge of the secret basis B , it is difficult to mimic the shortness of s .

We present these ideas concretely in algorithm 1.

Algorithm 1

The Hash-and-Sign Paradigm.

Public key a $n \times n$ non singular matrix A

Secret key a $n \times n$ non singular matrix B (well-conditioned)

Sign($m; H, B$)

Require A message $m \in \mathcal{M}$ and a hash function $H: \mathcal{M} \rightarrow \mathbb{R}^n$

Output a signature s such that $s + H(m) \in \mathcal{L}(B)$ and $\|s\|$ is small

Procedure

1. Compute $t = H(m)$

return $s = \lfloor tB^{-1} \rfloor B - t$.

Verify($s, m; H, A$)

Require A message $m \in \mathcal{M}$, the same hash function $H: \mathcal{M} \rightarrow \mathbb{R}^n$, and the signature $s \in \mathbb{R}^n$

Output Accept or Reject

Procedure

1. Compute $t = H(m)$

if $\|s\|$ is large or $t + s \notin \mathcal{L}(A)$

return Reject

else

return Accept

end if

2. By unit we mean the sides of this parallelepiped have the same length as the rows of B .

We make the following assumption about signatures produced by this paradigm.

Assumption 1 *Let $s_1, \dots, s_N$ be outputs of the hash-and-sign method in algorithm 1 on messages $m_1, \dots, m_N$. Then the points $s_1, \dots, s_N$ are spread uniformly over $[-\frac{1}{2}, \frac{1}{2}]^n B$.*

This assumption is based on the correctness of the hash function H . Heuristically, suppose the signatures formed non-uniform clusters in $[-\frac{1}{2}, \frac{1}{2}]^n B$. This would mean that the hashed messages which lie in some translated fundamental domain favour some corners of the domain to others. But this would undermine the uniformity of the output of the Hash function in $\mathbb{R}^n$.

3 Two Problems — HPP and HCP

Assumption 1 helps us extract an abstract problem that encapsulates a transcript attack on the hash-and-sign schemes. For this, let us begin by formally defining a parallelepiped.

Definition 1 Let B be an $n \times k$ real matrix with full column rank. Then we define the parallelepiped generated by B as the set

$$\mathcal{P}(B) = \{x B \mid -1 \leq x_i \leq 1\} := [-1, 1]^n B.$$

With this, defining a uniform distribution over $\mathcal{P}(B)$ follows immediately.

Definition 2 We say $y \sim \text{Uniform } \mathcal{P}(B)$ if $y = x B$ where $x_i \sim \text{Uniform}[-1, 1]$.

Here we go ahead and define the problem of our interest.

Problem 1 Hidden Parallelopiped Problem (HPP). Given N samples

$y_1, \dots, y_N$ drawn from the distribution Uniform $\mathcal{P}(B)$, approximate some row of B upto scalar multiple of -1 .

A special version of this problem is the following.

Problem 2 Hidden Cube Problem (HCP). Given N samples drawn from the distribution Uniform($\mathcal{P}(B)$), where B has orthonormal rows, approximate the rows of B upto scalar multiple -1 .

Remark 1 In assumption 1 we assumed that signatures from a hash-and-sign scheme are uniformly distributed over $[-\frac{1}{2}, \frac{1}{2}]^n B$. However we have defined the above problems over the set $[-1, 1]B$. This is not an issue because when $x \sim \text{Uniform}[-\frac{1}{2}, \frac{1}{2}]$, then $2x$ is Uniform $[-1, 1]$. So the samples from one can be converted to samples from the other. The alteration is made to make the calculations neater.

Our next lemma claims that an instance of a HPP can be converted to an instance of a HCP.

Lemma 1 Suppose $y \sim \text{Uniform } \mathcal{P}(B)$. Then there exists a matrix L such that $yL \sim \text{Uniform } \mathcal{P}(C)$ where C is a unit cube.

Proof Suppose $\Lambda = \mathcal{L}(B)$ where B is nonsingular. Let L be a matrix such that $LL^T = (B^TB)^{-1}$. Then

$$(BL)(BL)^T = B(B^TB)^{-1}B^T = I.$$

Hence the rows of BL are edges of a unit cube. If y is a uniform sample from $\mathcal{P}(B)$ then yL is a uniform sample from $\mathcal{P}(BL)$ where BL is a unit cube. $\square$

If we have $G = (B^TB)^{-1}$ we can compute L as the Cholesky factor of G . This L is only one such matrix with this property. Any LQ , where Q is an orthogonal matrix, will also satisfy this. However, let us recall that an attacker does not have access to the basis B , but only to uniform samples from $\mathcal{P}(B)$. But one observes that a function of B can be recovered from the samples as follows.

Lemma 2 *Let $y \sim \text{Uniform } \mathcal{P}(B)$. Then $3\mathbb{E}y^T y = B^T B$.*

Proof If $y \sim \text{Uniform } \mathcal{P}(B)$, then $y = xB$ where $x \sim \text{Uniform } [-1, 1]^n$. So

$$y^T y = B^T (x^T x) B.$$

Let $x = [x_1 \ \cdots \ x_n]$ where $x_i \sim \text{Uniform } [-1, 1]$ independently. Then note that $[x^T x]_{ij} = x_i x_j$. When $i = j$,

$$\mathbb{E}x_i^2 = \frac{1}{2} \int_{-1}^1 \gamma^2 d\gamma = \frac{1}{3}.$$

When $i \neq j$, we have $\mathbb{E}x_i x_j = \mathbb{E}x_i \mathbb{E}x_j = 0$. So, we get $\mathbb{E}x^T x = \frac{I}{3}$. This gives the required result. $\square$

Thus we have established that HPP can easily be reduced to the HCP (without access to the distribution). So, from this point onwards, we focus only on HCP.

4 Solution to the Hidden Cube Problem (HCP)

Let us restate HCP for our convenience.

Problem Hidden Cube Problem (HCP). Given N samples drawn from the distribution $y \sim \text{Uniform}(\mathcal{P}(B))$, where B has orthonormal rows, approximate the rows of B upto scalar multiple -1 .

The ingenuity of the discussed work [5] lies in looking at the fourth moment of y as an operator instead of a fixed number. This, as we will see, gives a simple characterization for the rows of B as points that minimize this operator. We begin by defining

$$m_{B,k}(w) := \mathbb{E}_y \langle y, w \rangle^k \quad \text{where} \quad y \sim \text{Uniform } \mathcal{P}(B).$$

Now, we will find an expression for $m_{B,4}(w)$.

$$\begin{aligned}
m_{B,4}(w) &= \mathbb{E} \langle y, w \rangle^4 \\
&= \mathbb{E} \left[\sum_{i=1}^n x_i \langle b_i, w \rangle \right]^4 \\
&= \sum_i \langle b_i, w \rangle^4 \mathbb{E} x_i^4 + 3 \sum_{i \neq j} \langle b_i, w \rangle^2 \langle b_j, w \rangle^2 \mathbb{E} x_i^2 \mathbb{E} x_j^2 \\
&= \frac{1}{5} \sum_i \langle b_i, w \rangle^4 + \frac{1}{3} \sum_{i \neq j} \langle b_i, w \rangle^2 \langle b_j, w \rangle^2 \\
&= \frac{1}{3} \sum_{i,j} \langle b_i, w \rangle^2 \langle b_j, w \rangle^2 + \left(\frac{1}{5} - \frac{1}{3} \right) \sum_i \langle b_i, w \rangle^4 \\
&= \frac{1}{3} \left(w \left[\sum_i b_i^T b_i \right] w^T \right)^2 - \frac{2}{15} \sum_i \langle b_i, w \rangle^4 \\
&= \frac{1}{3} (w B^T B w^T)^2 - \frac{2}{15} \sum_i \langle b_i, w \rangle^4 \\
&= \frac{1}{3} \|w\|^4 - \frac{2}{15} \sum_i \langle b_i, w \rangle^4.
\end{aligned}$$

The above set of equalities are elementary but some involve minor subtleties. For example, in the third inequality, expanding the fourth power will also have terms of the kind $x_i^3 x_j$, $x_i^2 x_j x_k$, $x_i x_j x_k x_l$. However since $\mathbb{E} x_i = 0$, all these terms will not contribute anything. Further, expanding the fourth power with the multinomial theorem would make the $x_i^2 x_j^2$ terms appear with a factor of 6. But in this case the i, j is an unordered pair. The above summation distinguishes $x_i x_j$ from $x_j x_i$, which cuts the factor down to 3. The second last equality is a standard exercise in linear algebra.

Finally, we want to solve the following constrained optimization problem.

Problem 3 Minimize $m_{B,4}(w)$ subject to $\|w\|^2 = 1$.

Solution Since b_i s are orthonormal and w is of unit size we have $|\langle b_i, w \rangle| \leq 1$. Hence, $\langle b_i, w \rangle^4 \leq \langle b_i, w \rangle^2$, or,

$$\sum_i \langle b_i, w \rangle^4 \leq \sum_i \langle b_i, w \rangle^2 = 1.$$

Immediately, $w = \pm b_i$ for some $i = 1, \dots, n$ achieves equality in the above inequality.

Furthermore, if $w \neq \pm b_i$ for any i , then $\langle b_i, w \rangle^4 < \langle b_i, w \rangle^2$ for some i , and hence the above inequality is strict. Since minimizing $m_{B,4}(w)$ is equivalent to maximizing the above sum, we get

$$\min_{\|w\|=1} m_{B,4}(w) = \frac{1}{3} - \frac{2}{15} = \frac{1}{5},$$

and the minimum is achieved in the points $\{\pm b_i\}_i$.

In practice, we do not have B and, therefore, not $m_{B,4}(w)$ either. But at any point w we can estimate $m_{B,4}(w)$ using the samples from Uniform $\mathcal{P}(B)$. This allows us a gradient based optimization method to minimize $m_{B,4}(w)$. To satisfy the constraint we either rewrite $m_{B,4}(w) = m_{B,4}(\theta)$ where θ , the angle vector represents a unit vector. Then we optimize over θ . But this will require extra calculations while approximating the moment function from the samples.

Alternatively, we can optimize over w , and normalize after every step, to drag the flow back to the unit sphere. The authors of [5] report the success of the second approach. Note that the gradient of $m_{B,4}(w)$ can be calculated as follows

$$\begin{aligned} & m_{B,4}(w + th) \\ &= \frac{1}{3} \langle w + th, w + th \rangle^2 - \frac{2}{15} \sum_i (\langle b_i, w \rangle + t \langle b_i, h \rangle)^4 \\ &= \frac{1}{3} [\langle w, w \rangle + 2t \langle w, h \rangle]^2 - \frac{2}{15} \sum_i [\langle b_i, w \rangle^4 + 4t \langle b_i, w \rangle^3 \langle b_i, h \rangle] + t^2(\dots) \\ &= \frac{1}{3} \|w\|^4 - \frac{2}{15} \sum_i \langle b_i, w \rangle^4 + t \left[\frac{4}{3} \|w\|^2 \langle w, h \rangle - \frac{8}{15} \sum_i \langle b_i, w \rangle^3 \langle b_i, h \rangle \right] + t^2(\dots). \end{aligned}$$

This shows that

$$\nabla m_{B,4}(w) = \frac{4}{3} \|w\|^2 w - \frac{8}{15} \sum_i \langle b_i, w \rangle^3 b_i.$$

Although we will not use this expression of the gradient in the algorithm (as we do not know b_i s), this may still be useful to analyze the rate of convergence

and decide an appropriate step size parameter. In practice we note that

$$\langle y, w + th \rangle^4 = \langle y, w \rangle^4 + t(4\langle y, w \rangle^3 \langle y, h \rangle) + t^2(\dots).$$

This gives us

$$\nabla m_{B,4}(w) = \nabla \mathbb{E} \langle y, w \rangle^4 = \mathbb{E} \nabla \langle y, w \rangle^4 = 4\mathbb{E} \langle y, w \rangle^3 y. \quad (1)$$

This is what we use to estimate the gradient from the samples. Now we explicitly state the algorithm discussed above.

Algorithm 2

Solution to HCP.

Parameters δ (step size for gradient descent)

Input Points $s_1, \dots, s_N$ sampled from Uniform $\mathcal{P}(B)$

Output $\pm b_i$ for some i

(let $m = m_{B,4}$)

Procedure

1. Pick w randomly such that $\|w\| = 1$.
 2. Use equation 1 to compute g as the approximation of $\nabla m(w)$.
 3. Let $w' = w - \delta g$.
 4. Normalize w' . That is $w' \leftarrow \frac{w'}{\|w'\|}$.
- if** $m(w') \geq m(w)$ [No reduction was made, termination condition. These moments are approximated using the samples.]
- then**
- return** w
- else**
- replace w by w' and return to step 2.
- end if**
-

Finally we revise the key steps of this exposition in the following steps. We recall that our goal is to approximate some row of B given signatures that are uniformly distributed over $[-\frac{1}{2}, \frac{1}{2}]^n B$.

1. We first convert the samples z from $[-\frac{1}{2}, \frac{1}{2}]^n B$ to samples x from $\mathcal{P}(B)$.
2. We transform these samples to approximate samples $y = xL$ from $\mathcal{P}(BL)$ where $LL^T = (3\mathbb{E}xx^T)^{-1}$. Here BL is orthogonal.
3. Finally we optimize the function $m_{B,4}(y)$ by using algorithm 2. The point of optima is approximately a row of B upto a negative multiple.

5 Further Discussion

Finally we mention some directions that one might seek to investigate. The above attack reveals a particular row of the secret basis. Can one extend the algorithm to find all the rows? This could be done by adding an regularizer to the objective that penalises solutions in the direction of vectors already found. In such a case the regularizer could be the norm of the projection of w onto the span of the recovered vectors. Or one may be able to project the samples to a space orthogonal to the span of already leaked vectors, thus looking at a lower dimensional problem. Empirically testing these approaches could be fruitful. We note here that in some signature schemes, like NTRUSign, the basis vectors have a special form [3]. In such cases, finding one basis vector may leak information about the other basis vectors as well.

In subsequent cryptographic schemes Babai's nearest plane algorithm [1] is used instead of the round-off method. In this case the signatures lie in $\mathcal{P}(\tilde{B})$ where $\tilde{B}$ is the Gram-Schmidt factor of B . The attack proposed in this note will recover rows of $\tilde{B}$. Can the attack be extended to recover B ? Can $\tilde{B}$ itself be exploited in a meaningful way?

Assumption 1 is the key factor in this attack. Later cryptographic schemes have been modified to primarily nullify this assumption. For example, subsequent frameworks use a variant of Klein's randomized nearest plane algorithm where the nearest lattice points are not sampled uniformly but from a Gaussian distribution around the target point [4]. This prevents a transcript of signatures from leaking the shape of the secret basis.

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